

Lesson 25: The Diffusion Equation - Separation of Variables

Professor Dave Explains Differential Equations Master Series · Lesson 25

Video: The Diffusion Equation Part 1: Separation of Variables Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that while the wave equation is reversible and oscillatory (u_{tt}), the Diffusion (or Heat) Equation $u_t = \alpha^2 u_{xx}$ is irreversible and dissipative. Heat flows from hot to cold, inexorably smoothing out sharp temperature spikes like milk diffusing into coffee. By separating variables into $u(x, t) = X(x)T(t)$, we see that higher spatial frequencies (rough, jagged gradients) decay exponentially faster in time ($\sim e^{-n^2 \pi^2 \alpha^2 t / L^2}$), leaving only the smooth fundamental profile.

2. Core Theoretical Framework & Rigorous Formulations

1. The Heat Conduction Equation

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}, \quad \alpha^2 = \frac{k}{\rho c_p} \quad (\text{Thermal Diffusivity})$$

Boundary conditions: $u(0, t) = 0, u(L, t) = 0$. Initial profile: $u(x, 0) = f(x)$.

2. Separation of Variables & Exponential Decay

$$u(x, t) = X(x)T(t) \implies \frac{T'(t)}{\alpha^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

- Spatial Modes: $X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2$
- Temporal Decay: $T_n(t) = \exp(-\alpha^2 \lambda_n t) = \exp\left(-\frac{n^2 \pi^2 \alpha^2}{L^2} t\right)$
- Full Solution: $u(x, t) = \sum_{n=1}^{\infty} c_n \exp\left(-\frac{n^2 \pi^2 \alpha^2}{L^2} t\right) \sin\left(\frac{n\pi x}{L}\right)$

3. Fully Solved Whiteboard Problems

Thermal Engineering Problem: Heat Dissipation in a Chilled Rod

Problem Statement: A metal bar of length $L = \pi$ with thermal diffusivity $\alpha^2 = 1$ has ends submerged in ice water ($u(0, t) = 0, u(\pi, t) = 0$). If initial temperature is $f(x) = 50 \sin(x) - 20 \sin(3x)$, find the temperature $u(x, t)$ for all $t > 0$.

Step 1: Check Boundary Eigenmodes

Length $L = \pi$, so $\sin(n\pi x/L) = \sin(nx)$, and decay rate is $\alpha^2 \lambda_n = (1)(n^2) = n^2$.

Step 2: Inspect Initial Condition for Orthogonal Harmonics

$u(x, 0) = 50 \sin(x) - 20 \sin(3x)$. Notice this is ALREADY an exact Fourier sine series! No integration required!

Step 3: Extract Fourier Coefficients Directly

$c_1 = 50, c_3 = -20$, and $c_n = 0$ for all other n .

Step 4: Attach Specific Temporal Decay Factors

For $n = 1$: decay factor is $e^{-1^2 t} = e^{-t}$. For $n = 3$: decay factor is $e^{-3^2 t} = e^{-9t}$.

Step 5: Assemble Complete Solution

$u(x, t) = 50e^{-t} \sin(x) - 20e^{-9t} \sin(3x)$.

Step 6: Physical Dissipation Analysis

The 3rd harmonic $\sin(3x)$ decays at rate e^{-9t} , which is 9 times faster than the fundamental mode e^{-t} ! For $t > 0.5$, $e^{-9(0.5)} = e^{-4.5} \approx 0.01$, so the 3rd harmonic has completely vanished, leaving only a pure smooth sine curve!

⚠ Common Exam Pitfall & Trapdoor Avoidance

Do not compute Fourier integrals $\int f(x) \sin(nx) dx$ if the initial condition is already expressed as a sum of sines. Simply read off the coefficients directly.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

The diffusion equation acts as an exponential low-pass filter: high-frequency spatial wiggles (n^2) vanish almost instantly.

- Video reference: [The Diffusion Equation Part 1: Separation of Variables](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.